A Centralized AI Operations Platform

Practical Technical Guide & System Overview

AI Health Check

ARTI-409-A · AI Systems & Governance · Capstone

"One schema. One audit chain.
One human-in-the-loop contract."

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April 2026

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1. Introduction

What this system is

The AI Health Check Platform is a single web application that gives a team one place to watch every AI service they run in production. Instead of piecing together five different tools to monitor quality, investigate problems, and prove compliance, the team uses one app that covers the full lifecycle.

Think of it as a control room for a fleet of AI services. Each service is a product that uses a language model under the hood. The platform keeps a live view of how those products are behaving, spots when something goes wrong, and guides the team through fixing it — all while keeping a tamper-evident record that an auditor can trust.

Who it is for

- Engineering teams running AI features in production and needing daily visibility.
- Technical leads responsible for AI quality and incident response.
- Compliance, governance, and audit teams who need evidence trails.
- Non-technical stakeholders who want to understand AI system health at a glance.

Why it is important

AI systems fail in ways that traditional monitoring does not catch. Servers can be 100% up while the model gives wrong answers. Users notice before anyone else. Investigations happen in chat threads and disappear. Audit evidence is scattered across screenshots and exports.

This platform closes all four of those gaps with one application, one database, and one approval contract that covers every AI action.

2. Problem We Are Solving

Why AI systems fail in real organizations

AI-powered products behave differently from normal software. The code may be healthy and the servers may be reachable, yet the actual answers the AI gives can silently decay. The team only finds out when a user complains or a report looks wrong.

The four chronic gaps

- **No shared runtime view —** teams track whether the server is up but cannot see whether the AI is behaving. Uptime is not behaviour.
- **Drift is caught by users, not tools —** answer quality slowly gets worse over weeks. Without a detector, the first signal is an angry customer.
- **Investigations live in chat —** when something breaks, the team diagnoses it in Slack, and the knowledge disappears. No tracked workflow, no permanent record.
- **Compliance evidence is stitched from many tools —** audit-time means exporting from five products and hoping nothing is missing. Silent gaps are the failure mode regulators care most about.

> **The short version**
> Traditional monitoring tells you the system is reachable. It does not tell you the AI is giving good answers, that humans reviewed every AI decision, or that the record is tamper-proof. That is the gap this platform closes.

3. Solution Overview (Big Picture)

What the platform does

The platform brings four concerns — registering services, monitoring them, investigating incidents, and proving compliance — into one application. Every action goes through the same approval logic, writes to the same audit chain, and is stored in the same database.

How it solves the problem

- **One registry —** every AI service has one record that owns its metadata, owner, environment, and sensitivity label.
- **Continuous evaluation —** the platform regularly tests every service with a small exam and scores the results. Quality is measured, not guessed.
- **Structured incidents —** when a problem appears, a tracked incident is opened, the AI drafts a summary, and a human reviewer approves it before it becomes official.
- **Tamper-evident log —** every action creates an audit row that is cryptographically linked to the one before it. Any alteration is detectable.

Key idea: the AI Operations Control Room

The platform is a single-pane control room. An operator glances at the Dashboard, sees traffic-light-style indicators across the entire AI fleet, and clicks into any red item to investigate. Every step they take — registering a service, running an evaluation, approving an AI-drafted summary, exporting a compliance report — happens in the same app and leaves a signed receipt.

> **One-line thesis**
> One schema, one audit chain, one human-in-the-loop contract — across monitoring, evaluation, triage, and governance.

4. System Overview (Simple Architecture)

The platform has four layers. Each layer does one job. A user request flows from the top down and a response flows back up.

The four layers

Frontend — the user interface

A modern web application built with React. It is what the team actually sees and clicks. Nine pages cover every module of the system: Dashboard, Services, Evaluations, Incidents, Governance, Settings, and so on. The interface is designed for a glance — numbers and charts update automatically without needing to refresh.

Backend — the logic and rules

A Python application built with FastAPI. It receives requests from the frontend, checks who is allowed to do what, runs the business logic, and writes to the database. All AI calls are routed through one central function, so every safety rule is enforced in one place.

Database — the storage

A SQLite database keeps every record: services, evaluations, incidents, maintenance plans, audit entries, and users. The audit log is append-only, protected by database triggers so rows cannot be altered or deleted.

LLM integration — the AI brain

The platform talks to Anthropic's Claude through one pipeline. Claude does the thinking for judge scoring, incident summaries, and compliance narratives. Every call passes through six safety stages before and after, and every call is fully logged.

How a request flows (simple example)

1. The user clicks a button in the browser.
2. The frontend sends a request to the backend over HTTPS.
3. The backend checks the user's role and, if needed, calls Claude through the safe pipeline.
4. The backend writes the result and an audit entry to the database.
5. The backend sends the result back to the frontend.
6. The frontend updates the screen.

> **One rule worth remembering**

> Every crossing between layers is audited. You cannot move data through this system without a record of who moved it and why.

5. Core Modules

The platform is built from four modules. Each one handles a specific part of the AI lifecycle. They share the same database and the same approval rules, so they work together naturally.

Module 1 — AI Service Registry

What it does

The registry is the phone book of the platform. Every AI service is registered once with its metadata: name, owner, environment (production, staging, development), the model it uses, its endpoint URL, and most importantly its sensitivity label (public, internal, or confidential).

Key features

- Create, read, update, and delete AI service records.
- Connection testing — the Ping button verifies that the endpoint is reachable and responsive.
- SSRF protection on every URL — the system refuses to test services that point to internal or dangerous addresses.
- Sensitivity labels that every downstream module respects.
- Search, filter, and view toggles for working with many services at once.

Why it matters

Without a registry, each module would need its own list of services and its own rules. With one registry, every other module (monitoring, incidents, governance) simply references a service by ID. Sensitivity labels propagate automatically, so a confidential service is treated with extra care everywhere it appears.

Module 2 — Monitoring & Evaluation

What metrics are tracked

- Latency — how long each call to a service takes (typical, slow, and worst values).
- Quality — how accurate and helpful the responses are (scored from 0 to 100).
- Error rate — how often quality falls below the acceptable threshold.
- Active alerts — problems that currently need attention.

What evaluation means

Evaluation means running a small exam against an AI service. The exam is a question bank of test cases — each with a question and an expected answer. The service replies to each question, and a separate AI judge scores the reply for factuality and hallucination. The run produces one overall quality score.

What drift detection is (simple explanation)

Drift is when quality gets worse over time. The platform detects drift in two ways. First, it checks whether the latest score dropped below a threshold (a sudden collapse). Second, it looks at a running window of recent scores and asks whether they are trending downward (a slow decay). Either kind raises an alert and shows up on the Dashboard.

> **Why two drift checks**
> Quality can fail in two ways. It can crash suddenly (a prompt got corrupted) or it can slide slowly over weeks. A threshold catches the first. A trend catches the second. A real platform needs both.

Module 3 — Incident & Maintenance

What happens when something goes wrong

When a drift alert fires, the operator clicks Create incident from the notification. An incident form opens, already pre-filled with the service, the severity, the symptoms, and a suggested triage category. The user confirms the details and submits. The incident is now a tracked record in the platform.

Incident creation and investigation

- Every incident has a lifecycle: Open, Investigating, Resolved, Closed.
- A triage checklist helps the operator categorise the likely cause (data issue, prompt change, model update, infrastructure, safety policy).
- The incident detail page has three zones: symptoms and checklist on the left, an AI-drafted stakeholder summary on the right, and a list of maintenance plans below.

LLM assistance with human approval

The platform can generate a draft summary of the incident using Claude. The checked triage boxes are fed to the AI so the summary is grounded in the operator's hypotheses. The draft appears with an orange banner that reads "Draft needs human approval." The operator reads the draft, types a reviewer note of at least 20 characters explaining what they verified, and clicks Approve. Only then does the draft become an official summary.

Maintenance planning

Once the cause is known, the team plans the fix. A maintenance plan records the risk level, the rollback strategy, the validation steps, and the scheduled date. A plan also needs explicit approval before it counts as agreed. The same approval pattern is used everywhere AI content or risky changes need to be signed off.

> **Human-in-the-loop contract**
> The AI drafts. The human approves with a note. Every time. The AI never publishes a summary, approves a plan, or closes an incident on its own.

Module 4 — Governance & Compliance

Role-based access

Three roles cover every user. Admins can do everything, including changing roles and overriding confidentiality gates. Maintainers can do most operational work but cannot change roles or send confidential data to the AI. Viewers have read-only access.

- Admin — full control and governance authority.
- Maintainer — can register services, run evaluations, manage incidents.
- Viewer — can see everything but cannot change anything.
- Every denied request is itself logged, so attempted misuse is on the record.

Audit logs

Every action anyone takes — every login, every service change, every incident approval, every failed attempt — writes a row to the audit log. The log is searchable and filterable, and it cannot be edited or deleted.

Data policy

Confidential services cannot have their data sent to the external AI without an explicit admin override, and that override is itself audited. The system has a safety scanner that blocks prompts containing injection patterns, detected PII, or toxic content. Both the input and the output of every AI call are scanned.

Compliance report export

An admin can pick a date range and export the whole story as a PDF or JSON report. The export bundles audit entries, incidents with their approvals and reviewer notes, and maintenance plans with their scheduled dates. Large exports include a warnings section that flags if any part was truncated — no silent gaps.

The integrity check

A single button verifies the entire audit log. The system walks every row, recomputes its cryptographic fingerprint, and confirms that every row is linked to the one before it. If anyone tampered with a row, the chain breaks at the exact point of tampering. Green banner means the record is trustworthy.

6. End-to-End Workflow

Here is the full story of the platform from a healthy starting point to a closed-out incident with an auditor-ready report. Every step uses modules and patterns introduced earlier.

1. Add the AI service. An admin opens the Services page, fills out a short form, and clicks Save. The URL is checked for safety, the service is registered, and an audit entry is written.
2. Run monitoring automatically. Every few minutes the platform pings the service and records its latency and health. The Dashboard shows the live values.
3. Evaluate performance on a schedule. Every hour the platform runs its test exam against each non-confidential service, scores the answers using an AI judge, and averages the results into one quality score.
4. Detect a problem. The new score drops below the acceptable threshold. The platform marks the run as drift and records an alert row in the same transaction.
5. Alert appears. Within 30 seconds the notification bell lights up and the Dashboard shows the drift in red.
6. Create an incident from the alert. The operator clicks Create incident on the alert. The incident form opens with all the fields already filled from the alert details.
7. Investigate with the checklist. The operator ticks the triage categories they think are most likely (for example, data issue and prompt change) and clicks Generate draft. The AI reads the symptoms and the checklist and writes a draft stakeholder summary.
8. Human approves the draft. The operator reads the draft, types a reviewer note of at least 20 characters explaining what they verified, and clicks Approve. The draft becomes the official summary with the operator's name and note attached.
9. Plan the maintenance. The team adds a plan with the risk level, rollback, validation steps, and scheduled date. The plan is then formally approved.
10. Log every step. From the first click to the final approval, every action has been writing to the audit chain. Nothing is lost.

Then the auditor arrives

1. The auditor gives a date range.
2. The admin opens Governance, picks the range, and clicks Export as PDF.
3. The admin also clicks Verify integrity. A green banner confirms the record is intact.
4. The admin hands the PDF and a screenshot of the green banner to the auditor. That is the artifact.

Why this is powerful

The whole story — from the failure to the fix — is one continuous record in one app. There is no switching between tools, no copying screenshots, and no gaps for the auditor to worry about.

7. How AI (LLM) is Used Safely

Where AI is used

- Judge scoring — a second AI call grades the answers from a service under test.
- Hallucination detection — a separate AI call checks whether the service fabricated anything.
- Incident summaries — AI drafts a stakeholder-friendly write-up of the problem and root causes.
- Dashboard insights — AI writes short, plain-language insights for trends the team should notice.
- Compliance reports — AI drafts the narrative section of a compliance export.

What AI helps with

AI speeds up reading, summarising, and drafting. It scores an answer in seconds, writes a stakeholder update in under a minute, and explains a compliance posture in plain language. None of this replaces the team's judgement — it just removes the blank page.

Why human approval is required

AI can hallucinate, misinterpret context, or generate plausible-sounding but wrong content. Every AI-generated artifact that goes into the record (incident summaries, compliance narratives, dashboard insights) must be reviewed by a named human. Approval requires a short note explaining what the reviewer verified. The draft is never promoted to official without that step.

The six-stage safety pipeline

Every AI call passes through six stages. Any stage can refuse the call before a single word reaches the model.

1. Sensitivity gate — confidential services require an admin override.
2. Input safety scan — the prompt is checked for injection patterns, PII, toxic content, and length.
3. Budget reserve — the call must fit inside the daily and per-minute budgets; concurrent calls cannot collectively overspend.
4. API call — the actual request to Claude, with automatic retry on transient failure.
5. Output safety scan — the response is checked for accidental PII before it leaves the backend.
6. Persist and audit — the call is saved with its cost, tokens, latency, and safety flags, and an audit entry is written.

> **The structural guarantee**
> There is exactly one place in the code that talks to the AI. Every feature that uses AI enters through

> that one function. Change a safety rule there, and every feature inherits it immediately.

8. Data Flow (Simple Explanation)

Data moves through the platform in a predictable path. Understanding that path helps explain where safety, audit, and approval checks sit.

Input → Processing → Storage → Display

1. Input — the user clicks a button in the web interface or fills out a form. The action is sent to the backend over HTTPS with an authentication token.
2. Processing — the backend validates the request, checks the user's role, applies safety rules, and (if needed) calls Claude through the safety pipeline.
3. Storage — the result is written to the database along with an audit entry. The audit entry is cryptographically linked to the previous one.
4. Display — the backend sends the result back to the frontend, which updates the screen. Dashboards update automatically on a refresh cycle.

Example: the full flow of one incident approval

- The reviewer clicks Approve in the browser.
- A modal asks for a note of at least 20 characters.
- The reviewer types the note and submits.
- The backend checks the note length again (because client-side rules can be bypassed).
- The backend flips the draft to an official summary and records the reviewer, the timestamp, and the note.
- An audit entry is written. Its cryptographic fingerprint links to the previous entry.
- The page refreshes and shows a green "Approved by" banner.

> **Where the trust lives**
> Every important check happens in the backend, not just the browser. Anyone with curl can bypass the UI — so the real rules live below the UI.

9. Testing & Reliability

Basic testing approach

The platform has a suite of automated tests that runs every time code is changed. Each test is a small, fast check — “does this feature still behave correctly?” If any test fails, the build fails and the change cannot ship.

- Around 120 tests covering authentication, services, evaluations, incidents, governance, and the AI pipeline.
- Coverage is measured and must stay above a floor; currently around seventy percent of the code is exercised.
- Tests run in seconds because they use an in-memory database and a simulated AI client — no network calls, no cost.

Tests that prove the tricky parts

- **Concurrency test —** twenty simulated callers hit the budget reservation at the same moment; the platform must still never exceed the budget.
- **Integrity test —** the test tampers with the audit log, then checks that the verify function detects the break.
- **Parser test —** the strict judge-score parser is fed malformed AI responses; only clean numbers are accepted, everything else is marked as a refusal.
- **SSRF test —** dozens of URLs are fed through the safety validator to prove dangerous addresses are blocked and legitimate ones are allowed.

Error handling

Every user-facing action handles failure. A service that does not respond records a failed connection and keeps the dashboard honest. An AI call that fails retries automatically up to twice before surfacing a clear error. Budget exhaustion or safety refusals return a clean message, not a cryptic stack trace.

Why reliability matters here

This is a governance tool. If the record cannot be trusted, the tool has no value. Every claim the platform makes — “AI calls are budget-limited,” “the audit log is tamper-evident,” “refusals are not counted as failures” — is backed by a specific test. That is what turns a demo into something you can defend in front of an auditor.

10. Governance & Risk Handling

Data privacy

Confidential services are isolated from the AI by default. Sending confidential data to an external model requires an admin to pass an explicit override, and the override itself is recorded. The safety scanner checks every prompt for emails, phone numbers, and SSNs, and scans every response for accidental PII leakage before it reaches the user.

AI risks addressed

- **Hallucination —** every AI-generated artifact is a draft until a human reviews it and adds a note. The draft is never the official version.
- **Drift —** threshold and trend detection catch both sudden collapse and slow decay of quality.
- **Prompt injection —** known injection patterns are blocked at the input stage; the platform refuses to process them.
- **Data leakage —** output is scanned before it leaves the backend, so sensitive content is redacted or refused.
- **Abuse of privilege —** role-based access enforces who can do what. Every denied action is logged as an attempt.
- **Runaway spend —** daily and per-minute budgets are enforced atomically so even a surge of concurrent requests cannot exceed them.

How the system reduces risk

Defence in depth. No single safeguard carries the whole weight. The approval contract requires a reviewer note at three points (browser, request boundary, and server). The audit log is append-only both by design and by database trigger. The AI pipeline runs six safety stages. If any one defence fails, the others catch the rest.

11. Key Strengths of the System

- **Unified platform —** monitoring, evaluation, incidents, and governance in one application with one database.
- **Governance-first design —** audit trails and human approvals are not bolted on; they are the skeleton of the app.
- **Human-in-the-loop contract —** the AI drafts, the human approves with a note, every time. Same contract for incidents, dashboard insights, and compliance reports.
- **Tamper-evident audit —** every action is cryptographically chained. A single button proves the chain is intact.
- **Single AI entry point —** every AI call, from every feature, goes through one function. Safety rules are enforced in one place.
- **Concurrency-safe budgets —** even twenty simultaneous requests cannot collectively exceed the daily cap. Proven by a test.
- **Auditor-ready exports —** pick a date range, download a PDF, hand it over. No stitching, no screenshots, no silent truncation.
- **Defence in depth —** every critical rule is enforced in at least two layers. No single point of trust.
- **Honest scope —** the platform does not pretend to be a multi-tenant enterprise product. It is a reference implementation that maps cleanly onto production technologies.

12. Limitations (Honest Section)

This is a capstone reference implementation. It is built to show that the governance model works end to end, not to be a ready-to-deploy enterprise product. The following simplifications are deliberate.

- **SQLite, not Postgres —** the database is a single file. A real deployment would use Postgres with full migrations. The schema is portable — the swap is a configuration change.
- **In-process scheduler —** background jobs run inside the same process. A real deployment would use a distributed worker queue such as Celery or RQ. The platform boundary is a drop-in replacement.
- **Single AI judge —** one model grades every answer. A production system would use an ensemble of judges to reduce bias from any single model.
- **No multi-tenancy —** every user in the database is in one organisation. A multi-tenant system would add tenant scoping on every query.
- **No SSO or SAML —** authentication is username and password with lockout. A real deployment would integrate with a corporate identity provider.
- **Simulated data for the demo —** several services, evaluations, and drift scenarios are seeded so the demo is reliable. In a real system this data would grow organically over time.
- **AI costs are simulated locally —** token costs are computed from a pricing table. Real billing would reconcile against Anthropic's own invoice.
- **No row-level access control —** role-based access decides what actions a user can take, not which services they can see. Per-team isolation is a documented next step.

> **Why say this out loud**
> An honest limitations section earns more trust than a flawless demo. Every simplification here is a deliberate choice, every one has a clear next step, and none of them breaks the core thesis of the platform.

13. Future Improvements

The limitations above map directly onto a roadmap. Everything here is additive — the core design does not need to change.

Next three to six months

- Migrate the database from SQLite to PostgreSQL with versioned schema migrations.
- Move background jobs to a distributed worker queue (Celery with Redis).
- Add single sign-on through SAML or OIDC.
- Introduce per-team row-level access so organisations can isolate their services.
- Add an email and Slack notification channel for critical alerts.

Next year

- Multi-model judge ensemble to reduce bias and improve scoring reliability.
- HIPAA, SOX, and GDPR compliance templates for out-of-the-box regulated workloads.
- Mobile triage app for on-call engineers to review AI drafts from a phone.
- External timestamp anchoring of the audit chain to a write-once-read-many store.
- Ability to register services for non-Anthropic models (OpenAI, Gemini, self-hosted) through a provider-abstracted client.

How the system can scale

The architecture is ready for scale. The database and scheduler are the two pieces that need to swap out; every other component already works as it will in production. The one AI entry point means a single code change flips the system from Anthropic-only to multi-provider. Audit chain integrity costs the same at one million rows as at one thousand — verification is still linear.

14. Quick Summary (Revision Section)

Use these ten points for quick revision before the presentation or for a fast overview of the system.

1. The platform is a single web app that monitors AI services, evaluates their quality, investigates incidents, and produces compliance reports.
2. It solves four gaps: no runtime view, drift caught too late, investigations in chat, and compliance stitched from many tools.
3. Four modules: Service Registry, Monitoring and Evaluation, Incidents and Maintenance, Governance and Compliance.
4. Quality is measured by running a small exam against each service; a second AI call grades the answers; drift is detected two ways.
5. Every AI-generated artifact is a draft. A named human must approve it with a note of at least twenty characters before it becomes official.
6. Every AI call passes through six safety stages: sensitivity check, input scan, budget reserve, API call, output scan, and audit persist.
7. Every action writes to a cryptographically chained audit log; one button verifies the entire chain and reports any tampering.
8. Three roles govern access: Admin, Maintainer, Viewer. Every denied attempt is itself logged, so misuse is visible.
9. The platform is a reference implementation — SQLite and an in-process scheduler are deliberate simplifications that swap cleanly for production components.
10. The thesis in one line: one schema, one audit chain, one human-in-the-loop contract.

Bonus — How to Explain This System in 1–2 Minutes

For a presentation opening, an interview, or a quick answer in Q&A, this is the story in under two minutes.

> **The elevator version**
> We built a centralised AI operations platform. Teams running AI services in production have no single place to see how those services are behaving — uptime is tracked, behaviour is not. Our platform gives them one dashboard for quality, a structured incident workflow when something breaks, and a tamper-evident audit trail that an auditor can trust. Every AI-generated summary, draft, or insight must be reviewed and approved by a named human with a note. Every action is cryptographically chained. One schema, one audit chain, one human-in-the-loop contract. It is a reference implementation with 123 automated tests, so every claim the platform makes is backed by a test that proves it works.

Shorter versions

The thirty-second version

It is a single web app that lets a team monitor their AI services, catch drift when quality drops, investigate incidents with AI-drafted summaries that a human must approve, and export a tamper-evident compliance report an auditor can trust — all in one place with one audit chain.

The one-sentence version

A single-pane AI operations control room where every action is gated, every denial is audited, every AI draft needs a human to sign off, and every change is cryptographically chained.

For a Q&A answer

When asked “what does your platform actually do?” start with the problem: “Teams running AI features have no single place to see whether their AI is still behaving correctly. Uptime does not mean quality.” Then give the solution: “Our platform is one web app that covers monitoring, evaluation, incidents, and compliance — with a human-in-the-loop contract on every AI output and a tamper-evident audit chain.” Finish with the proof: “One hundred and twenty-three automated tests back the platform’s claims. A single button verifies the audit chain is intact. A single date range produces a compliance report an auditor can read.”

Points you can always fall back on

- One schema, one audit chain, one approval contract — across every module.
- The AI drafts. The human approves. Every time.
- Every claim is backed by a specific test.
- An auditor's view is one PDF and one verify button.
- Defence in depth — no single safety check carries the whole weight.